An External Audit of OpenAI's Quasi-Riemann Hypothesis Manuscript

Zero-Free Half-Plane $\mathrm{Re}(s) > 7/8$

Master audit, loss table and residual-risk notes

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AI-assisted, multi-agent audit · github.com/data1493/quasi-rh-audit

Abstract

This is an external audit of the existing manuscript, not an independent re-proof of the analytic estimates. Primary objects: OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\mathrm{Re}(s) > 7/8$ (preprint, 30 Sep 2026) [1] and its Lean formalization, `github.com/openai/math`, commit `adc7f12` [2]. Residual risk: the margins are small, and ruling out an unlisted intermediate loss larger than the reserved budgets requires a specialist reading of the original manuscript. Source and issue tracker: github.com/data1493/quasi-rh-audit.

At a glance

Date	2026-10-13
Audited source	OpenAI preprint (30 Sep 2026) and Lean formalization in `openai/math`, commit `adc7f12`; audited files byte-identical through `1581d4c`
Scope	Audit of the existing paper and its own exponent accounting. Not a re-proof. The Oct. 5 "alternate 11/12" preprint is not covered.
Method	Layer-by-layer reconstruction and stress-test using an AI-assisted multi-agent workflow (SuperGrok Heavy), led by data1493, reading the public TeX source and Lean formalization.
Lean	A third-party Lean kernel replay (davegoldblatt) is reported for the zeta statement [3]; we built no Lean ourselves.
Status	No human specialist has yet reviewed these findings.

Summary for non-specialists

OpenAI's paper claims that the Riemann zeta function has no zeros to the right of the line $\mathrm{Re}(s) = 7/8$. We did not try to prove that again. Instead we checked the paper's own bookkeeping: at each step the proof "spends" a little of a small safety margin, and we checked that the margin is still positive after every loss the paper lists. It is: $1/4800$ in Stage 1, and $49/440640 > 1/10000$ at the Stage 2 endpoint. The catch is that these margins are small. If some intermediate step hides a loss the paper did not list, the margin could be used up, and only a specialist reading can rule that out. That is the review we are asking for.

1. Claims (precise scope)

1. The logical architecture of the proof is coherent at every examined layer.
2. The paper's own exponent ledgers for the modest positive margins (Stage 1 $1/4800$, Stage 2 endpoint $49/440640$) close with a positive remainder under the losses they explicitly charge.
3. The Lean formalization of the final theorems aligns with the claimed statements; the final theorems are machine-checked, and an independent third-party kernel replay has been reported (see Part 9).
4. No structural gap or disproof has been identified.

What this is not. A complete independent proof of the analytic estimates. The residual risk is quantitative and localized to the possibility of an unlisted loss in an intermediate lemma.

2. Architecture (Parts 1–9)

Part	Layer	Finding
1	Continuation criterion	Dual representations produce contradiction if zero lies too far right. No structural gap found at this layer.
2	Cubic-theta reflection	Exact under stated conditions; retains zero-on-nonunit restrictions.
3	Poisson & principal signal	Principal $1/L$ Mellin signal cleanly isolated.
4	Large sieve & row energy	Specialized sextic sieve + envelope $R(\delta)$ supply the positive margins.
5	Zero detector & moments	Saturation forces inverse witness dominant; feeds the envelope.
6	Stage 1 assembly (11/12)	Balanced scales, $C(s) = s - 2/3$, margin $1/4800$ under stated accounting.
7	Stage 2 refinement (7/8)	Compensation + asymmetric scales, endpoint positive.
8	Uniformity & transfer	Margins independent of target; quadratic factorization transfers to Dirichlet L-functions.
9	Lean alignment	Final theorems formalized and kernel-checked; third-party replay reported.

3. Specialist quantitative audit

Stage 1 high-side saving

- Reserved: $m_0 = 1/1200$, $\epsilon_* = 10^{-6}$, $m = 1/2400$.
- Central losses: $46\epsilon_* < 1/4800$.
- Principal and small-row losses each $< m_0/4$.
- After fixed scalar and late-height closure: claimed remainder $1/4800$.
- Status: table closes with positive remainder under charged losses.

Stage 2 endpoint certificate

- Algebraic identity yields $-E_* \geq 49/440640 > 1/10000$.
- Additional losses charged to reserved fraction of high-side margin or arbitrarily small ϵ_{total}.
- Status: remainder positive for every $\Delta = \beta_* - 7/8 > 0$.

Computational confirmation

- Row envelope $R(\delta) = \min\{1, \max(1 - \delta/2, 4/3 - \delta)\}$ evaluated; strict saving for $\delta > 1/3$.
- Endpoint constant $49/440640 \approx 0.0001112 > 1/10000$ checked arithmetically (the algebraic identity behind it was not re-derived).

4. Logical hinge (prose outline)

No Lean skeleton was built for this audit; this is a prose outline of how the formal statements line up with the hinge. The continuation criterion has been isolated as a pure logical implication: dual bounds with uniform positive powers imply holomorphic continuation of $1/L$ past the supposed rightmost zero. The concrete margins appear as explicit constants. The analytic estimates that discharge the implication remain in the prose.

5. Specialist loss table

Stage 1 (11/12) and Stage 2 (7/8) endpoint certificates. Self-contained extraction of every charged loss that feeds the final positive margins, with the reserved budgets they are charged against.

5.1 Stage 1 — high-side saving $1/4800$

Source: Proposition on balanced-high estimate and Lemma on late-height-closure.

Table 1. Reserved parameters (Stage 1)

Parameter	Value	Role
m_0	$1/1200$	Minimum error-free saving on every major contribution
ϵ_*	10^{-6}	Unit loss charged to each listed class
m	$1/2400$	Saving used after preliminary losses
ω	$\Delta_1/2$	Low-bound room

Table 2. Charged losses (Stage 1). Notes [1]–[6] below.

Loss class	Coefficient	Charged	Budget
Central[1]	$13\epsilon_* + 2(15\epsilon_*) + 3\epsilon_*$	$46\epsilon_*$	$< m_0/4 = 1/4800$
Principal residues[2]	$\le (3/2+1)\epsilon_*, 2\epsilon_*$	each $< m_0/4$	$m_0/4$
Small-row	$\le 2\epsilon_*$	$< m_0/4$	$m_0/4$
Large-row	requested ϵ_*	absorbed	reserved
Height factor[3]	late choice of τ	absorbed[4]	late-height closure
Contour tails[5]	increase N after target	$\ll$ any power[6]	external order

[1] Central: row count, numerator, local correction, reciprocal, multiplicity, physical dyads.

[2] Principal residues: two terms.

[3] Height factor: $(1+T_1)^{A\eta}$.

[4] Absorbed into half the remaining margin.

[5] Contour tails: $O(Z^{B\eta}T_1^{-N})$.

[6] Smaller than any positive power.

Accounting.

- The listed charges are tiny in absolute terms (central $46\epsilon_*$, principal $\le 2.5\epsilon_*$ and $2\epsilon_*$, small-row $\le 2\epsilon_*$; about 5.3×10^{-5} in total, against $m_0/4 \approx 2.08\times10^{-4}$). After all listed losses and division by the fixed scalar c_S, the paper retains at least $3m_0/4 = 1/1600$ before the height factor.

- Using only $m = 1/2400$ leaves an unused power $1/4800$ ($= 1/1600 - 1/2400$), which absorbs fixed constants from internal tail choices.
- Late-height closure absorbs the height factor and returns half of m.
- Final claimed high-side saving: $m/2 = 1/4800$ (paper, end of proof of the balanced-high estimate). The two appearances of $1/4800$ are different quantities that happen to be equal.

Status. Every listed loss is strictly smaller than its reserved fraction under the paper's accounting. The table closes with a positive remainder.

5.2 Stage 2 — endpoint certificate $49/440640$

Source: Lemma on balanced-endpoint (or equivalent high-bin endpoint calculation).

Certificate derivation (extracted). The high exponent at the endpoint satisfies an identity of the form

$$10368\,\mathcal{J}(-E_*) = (\text{non-negative quadratic form in } \delta, y).$$

The leading constant term is 49. Bounding the remaining factors ($\mathcal{J} \le 5/2$, $v \le 17(3+5y)$) produces

$$-E_* \;\ge\; \frac{49}{440640} \;>\; \frac{1}{10000}.$$

The inequality holds uniformly for $1/50 < \delta \le 5/6$.

Table 3. Charged losses against the Stage 2 certificate. Notes [7]–[10] below.

Loss class	Treatment	Budget
Amplitude bins[7]	absorbed into $O(\epsilon)$	reserved fraction[8]
Capacity reduction (ν_0)	charged $2q\nu_0 + \delta \max w_i$	$\Delta/4$ or equivalent
Height factor	late τ as in Stage 1	half high margin
Contour tails	external order N	$\ll$ any power[9]
Compensation[10]	absorbed into same ϵ_{total}	target-independent coeff.

[7] Amplitude bin width ϑ, witness subdivisions.

[8] Reserved fraction of high margin.

[9] Smaller than any positive power.

[10] Compensation and asymmetric dyadics; budget is a target-independent coefficient.

Accounting.

- The constant term alone already exceeds $1/10000$.
- All additional losses are charged to a reserved fraction of the high-side margin (proportional to $\Delta = \beta_* - 7/8$) or to an arbitrarily small ϵ_{total}.
- The certificate remains positive for every $\Delta > 0$.

Status. The endpoint inequality is negative by a fixed positive amount plus a term proportional to Δ. Under the paper's accounting the remainder stays positive.

5.3 Overall specialist conclusion

Both certificates close with a positive remainder under the losses the paper explicitly charges. The Stage 1 table accounts for every listed ϵ_* class and leaves $1/4800$. The Stage 2 certificate supplies a fixed positive constant larger than $1/10000$ that survives the same style of reserved-budget accounting. No listed loss exceeds its budget in the extracted text. The quantitative hinge survives this level of extraction.

Remaining limitation. The table uses the paper's own charged amounts. Full certification still requires confirming that no intermediate lemma contains an additional unlisted loss of larger size. Within the extracted accounting, the margins hold.

6. Residual risk

The margins are small. Within the extracted text, they survive every loss the paper explicitly charges and that the computational constants match. The residual risk is the existence of an unlisted intermediate loss larger than the reserved budget. Full certification requires specialist verification of every lemma that feeds the ledger, or formalization of those steps.

Where the risk sits

Editorial index compiled for this repository. Every statement below is taken from the audit files linked; nothing here adds a new finding.

The margins are small: a Stage 1 high-side saving of $1/4800$ and a Stage 2 endpoint constant $49/440640 \approx 0.0001112 > 1/10000$. Within the extracted text, these survive every loss the paper explicitly charges and that the computational constants match (00, 11, 17). The residual risk is the existence of an unlisted intermediate loss larger than the reserved budget. Confirming that none exists requires a specialist reading of every lemma that feeds the ledger, or formalization of those steps.

Item by item

Stage 1 central loss, $46\epsilon_* < m_0/4$ (file 12)
Additional loss classes in the full central-contribution equation not included in the 13 or the two 15s.

Late-height closure (file 13)
An estimate producing $A\eta$ that itself depends on a height not yet closed.

Specialized sextic large sieve, $(KD)^{2/3}$ (files 14, 04)
An extra factor in the full write-up not absorbed into ϵ; an error in the mod-6 recursion would degrade $R(\delta)$.

Zero-detector saturation (files 15, 05)
Product lower bound weaker than claimed.

Stage 2 compensation identities (files 16, 07)
An uncancelled, target-dependent main term in an intermediate expansion.

Uniformity in the conductor (files 08, 01)
A slowly growing conductor factor hidden inside an ϵ.

Lean translation fidelity (file 09)
Lean certifies the final theorems, not every intermediate prose estimate.

What would settle it

A short specialist note either confirming that the listed losses are fully accounted for under the paper's methods, or naming a specific lemma and loss that exceeds the reserved budget. See 18.

7. Conclusion

The manuscript survives structural scrutiny, Lean alignment, and specialist audit of its own exponent accounting. No disproof has been found. The quantitative hinge is localized and, under the stated accounting, closes with a positive remainder. The paper's own accounting is internally consistent under its methods; the result has not been independently re-proved from first principles.

Artifacts

- `01_continuation_criterion.md` – `09_lean_alignment.md`
- `10_specialist_margins.md`
- `11_loss_table.md`
- `17_computational_check.md`
- `19_residual_risk_notes.md`
- This document; source and issue tracker at `github.com/data1493/quasi-rh-audit`

References

[1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\mathrm{Re}(s) > 7/8$, preprint, 30 September 2026. Primary source.

[2] OpenAI, Lean formalization, `github.com/openai/math`, commit `adc7f12` (audited files byte-identical through `1581d4c`). Primary source.

[3] davegoldblatt, third-party Lean kernel replay of the zeta statement (reported).

[4] data1493, *External audit of OpenAI's quasi-Riemann hypothesis manuscript*, `github.com/data1493/quasi-rh-audit`, CC BY 4.0.